Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Plane shapes: types, properties and perimeter. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	2
Topic	Plane Shapes (2-D Shapes)
Subtopic	Plane shapes: types, properties and perimeter
Theme	—
Curriculum objectives	• identify the types of plane shapes and state their properties, similarities and differences • identify the parts of a circle • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle
Previous knowledge	• Students have been taught angles.
Lesson sequence	• identify the types of plane shapes and state their properties, similarities and differences — the teacher draws the various shapes on the board; the student identify their properties; the teacher writes the properties of the shapes on the board for the students to learn • identify the parts of a circle — the teacher explains the parts of a circle to the students: the arc, radius, chord and diameter, with the circumference, segment, sector and tangent labelled on the board drawings • find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle — the teacher works the perimeter of plane shapes on the board: square, rectangle, trapezium, parallelogram, regular hexagon and circle, including finding a missing side or radius when the perimeter is given
Instructional materials	• Cardboard containing angles presentation.
Evaluation	• Find the perimeter of a square of sides 8cm • Find the perimeter of a rectangle of length 7cm and breadth 4cm. • What is the perimeter of regular pentagon with sides 5cm. • How many triangles can an octagon be divided into if you draw diagonals from one vertex to all non-adjacent vertices? • Find the perimeter of a parallelogram whose base is 6cm and sides 5cm • The circumference of a circle is 22cm – Find the radius of a circle ($\pi = \frac{22}{7}$).

	• A rectangle is such that its length is twice its breadth, if the breadth is d, find the total length of the sides of rectangle. • Calculate the perimeter of the figure below. [the figure is not included here, marked 1cm, 5cm, 9cm] • Find the perimeter of a regular hexagon with sides 5cm.

Lesson objectives

1.	identify the types of plane shapes and state their properties, similarities and differences
2.	identify the parts of a circle
3.	find the perimeter of a regular polygon, square, rectangle, trapezium, parallelogram and circle

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Plane shapes: types, properties and perimeter

Class: JSS1 **Subject:** Mathematics **Week:** 2

Section 1: Meet the Plane Shapes: Types and Properties

A plane shape is a flat, two-dimensional (2-D) shape. It has length and width but no thickness, so it lies completely flat on a surface. Look around your desk: the top of your book, the face of a clock, and a floor tile are all plane shapes.

Every plane shape is named by the number of its straight sides, and each has its own set of properties — facts that are always true about it. Knowing these properties lets you tell one shape from another, even when two shapes look similar at a glance.

Here are the most common plane shapes and the properties that belong to each one.

The triangle has 3 sides and 3 angles. The angles always add up to 180°. A triangle is the only closed shape with just three straight sides.

The quadrilateral is a family of shapes that each have 4 sides and 4 angles. Their angles always add up to 360°. Four special quadrilaterals are worth knowing well:

• Square — all 4 sides are equal, and all 4 angles are right angles (90°). Its opposite sides are parallel.
• Rectangle — opposite sides are equal and parallel, and all 4 angles are right angles.
• Trapezium — has exactly one pair of parallel sides. Its other two sides are not parallel.
• Parallelogram — both pairs of opposite sides are parallel and equal, but its angles are not right angles.

The pentagon has 5 sides and 5 angles. The hexagon has 6 sides and 6 angles. The octagon has 8 sides and 8 angles. A stop sign is a good real-life octagon.

The circle is different from all the shapes above: it has no straight sides at all. It is a closed curve where every point on the curve is the same distance from its centre. That fixed distance is called the radius, and a line that passes right across the circle through the centre is called the diameter. The diameter is twice the radius.

Seeing similarities and differences between shapes is a handy skill. Two shapes may share a property, and one property alone is often not enough to identify a shape.

For example, a square and a rectangle are similar because both have 4 sides and 4 right angles. But they differ because a square has all four sides equal, while a rectangle only needs its opposite sides equal. So a rectangle is not a square unless all its sides happen to be equal.

A square and a parallelogram are similar because both have opposite sides parallel. But they differ because a square has four right angles while a parallelogram does not, and a square has all sides equal while a parallelogram only needs opposite sides equal.

A triangle and a quadrilateral both have straight sides and are both plane shapes, but they differ in the number of sides: a triangle has 3, a quadrilateral has 4.

Finally, the circle stands apart. Every other shape in this list is made of straight sides, but the circle is the only one with a curved boundary. That single difference is what separates a circle from all the polygons.

A good way to keep track is to ask two questions about any shape: How many sides does it have? and Are any of those sides parallel, equal, or at right angles? The answers together pick out the shape every time.

Worked example 1 — Sorting shapes by their properties

A set of shapes is shown in front of you: a square, a rectangle, a trapezium, a parallelogram, and a circle. Use their properties to sort them.

First, split them into "curved" and "straight-sided." Only the circle is curved, so it stands alone. The other four all have straight sides.

Now look at the straight-sided shapes. A trapezium has exactly one pair of parallel sides; a parallelogram has two pairs of parallel sides. So the parallelogram is the one with both pairs of opposite sides parallel.

Between the square and the rectangle, both have four right angles. The difference is that all four sides of the square are equal, while the rectangle only has its opposite sides equal. So the square is the equal-sided one with right angles, and the rectangle is the one with right angles but only opposite sides equal.

Answer: Circle (curved, no sides) — Trapezium (one pair of parallel sides) — Parallelogram (two pairs of parallel sides) — Rectangle (right angles, only opposite sides equal) — Square (right angles, all sides equal).

Worked example 2 — Comparing two shapes

State one similarity and one difference between a square and a parallelogram.

Similarity: Both a square and a parallelogram have opposite sides that are parallel.

Difference: A square has all four sides equal and all four angles are right angles, whereas a parallelogram has opposite sides equal but its angles are not right angles.

Answer: Similar — opposite sides are parallel. Different — the square has right angles and equal sides all round; the parallelogram has no right angles and only its opposite sides are equal.

Practice

1. Name the plane shape that has exactly one pair of parallel sides, and name the shape that has both pairs of opposite sides parallel but no right angles.
2. Write down one similarity and one difference between a square and a rectangle.
3. A shape has no straight sides and every point on its boundary is the same distance from its centre. What is it called, and what name is given to that fixed distance from the centre to the boundary?
4. How many sides and angles does a hexagon have, and how does that number compare with the sides of a triangle?

Section 2: Getting to Know the Circle

A circle is one of the plane shapes we see everywhere — a bicycle wheel, a clock face, a coin. Unlike a square or a rectangle, a circle has no straight sides and no corners. Instead, it is a perfectly round shape in which every point on the edge is the same distance from one fixed point at the very middle. That fixed middle point is called the centre.

Because a circle has no sides, we name and measure it using special parts. Let's meet the most important ones.

Centre — the exact middle point of the circle. It is usually marked with a letter such as O.

Radius — the straight line from the centre to any point on the edge of the circle. Every radius in the same circle is the same length. We often write the radius as r.

Diameter — a straight line that passes through the centre and touches the edge at both ends. The diameter is always twice as long as the radius: diameter = 2 × radius, or d = 2r.

Circumference — the distance all the way around the edge of the circle. It is the circle's version of a perimeter. We find it using the formula circumference = 2 × π × radius, written as C = 2πr (which is the same as C = πd, using the diameter).

Chord — a straight line joining two points on the edge of the circle, without necessarily passing through the centre. A diameter is actually a special chord — the longest one, because it passes through the centre.

Arc — a part of the edge (circumference) of the circle, like a curved section of the rim.

Sector — the region inside the circle between two radii and the arc they cut off. It looks like a slice of cake.

Segment — the region inside the circle between a chord and the arc it cuts off.

Tangent — a straight line that touches the circle at exactly one point and does not cut through it.

Here are the key links to remember:
- The radius is half the diameter: r = d ÷ 2.
- The diameter is double the radius: d = 2r.
- The circumference goes all the way around, and π (pi) is roughly 3.14 or 22/7.

Worked Example 1 — Naming the parts

Look at this list of descriptions and match each one to the correct part of a circle.

1. The distance around the edge of a circle.
2. A line from the centre to the edge.
3. A line through the centre touching the edge at both ends.

Solution.
1. The distance around the edge is the circumference.
2. A line from the centre to the edge is the radius.
3. A line through the centre touching the edge at both ends is the diameter.

So the answers are: 1 → circumference, 2 → radius, 3 → diameter.

Worked Example 2 — Radius and diameter

The radius of a bicycle wheel is 14 cm. What is the diameter?

Solution.
Remember that the diameter is twice the radius.

d = 2r
d = 2 × 14
d = 28 cm

So the diameter of the wheel is 28 cm.

Worked Example 3 — Finding the circumference

Find the circumference of a circle whose radius is 7 cm. (Use π = 22/7.)

Solution.
Use the formula C = 2πr.

C = 2 × (22/7) × 7
C = 2 × 22
C = 44 cm

So the circumference is 44 cm.

Practice

1. a. A circle has a radius of 9 cm. What is its diameter?
   b. A circle has a diameter of 30 cm. What is its radius?
   c. Which part of a circle is the distance all the way around the edge?
   d. A chord that passes through the centre of a circle is given a special name. What is it?
   e. Find the circumference of a circle with a radius of 10 cm. Use π = 3.14.
   f. Name the part of a circle that is a slice of cake shape — the region between two radii and the arc they cut off.

Section 3: Finding the Perimeter of Plane Shapes

The **perimeter** of a shape is the total distance around its outside edge. You find it by adding up the lengths of all its sides. For some shapes there are handy shortcuts, and for a circle the perimeter has a special name — the **circumference**. Here are the ways to find the perimeter of the shapes in this lesson. Let s stand for the length of one side of a shape.

Square. All four sides are equal, so the perimeter is $4 \times s$, or simply $4s$.

Rectangle. The two longer sides are equal and the two shorter sides are equal. If the rectangle is l long and w wide, the perimeter is $2l + 2w$, or $2(l + w)$.

Regular polygon. A regular polygon has all sides equal and all angles equal. A regular polygon with n sides, each of length s, has perimeter $n \times s$. For example, a regular pentagon has 5 equal sides, so its perimeter is $5s$.

Trapezium. A trapezium has four sides, and only the two opposite sides that are parallel are special — the other two sides may be different. So you just add all four side lengths together.

Parallelogram. Opposite sides are equal in length, so if the two different side lengths are a and b, the perimeter is $2a + 2b$, or $2(a + b)$.

Circle. The perimeter of a circle is called the **circumference**. It is found using the diameter d or the radius r: circumference $= \pi d$, or circumference $= 2\pi r$. We often use $\pi \approx \frac{22}{7}$ or $\pi \approx 3.14$.

Worked example 1 — a square and a rectangle. A square has a side of 6 cm. Its perimeter is $4 \times 6 = 24$ cm. A rectangle is 10 cm long and 4 cm wide. Its perimeter is $2(10 + 4) = 2 \times 14 = 28$ cm.

Worked example 2 — a regular polygon. A regular hexagon has 6 equal sides, each 5 cm long. Its perimeter is $6 \times 5 = 30$ cm.

Worked example 3 — a trapezium. A trapezium has sides of 7 cm, 4 cm, 9 cm and 5 cm. Add them all: $7 + 4 + 9 + 5 = 25$ cm.

Worked example 4 — a parallelogram. A parallelogram has sides of 8 cm and 3 cm. Its perimeter is $2(8 + 3) = 2 \times 11 = 22$ cm.

Worked example 5 — a circle. A circle has a radius of 7 cm. Its circumference is $2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm.

Practice

1. A square has a side of 9 m. What is its perimeter?
2. A rectangle is 12 cm long and 7 cm wide. Find its perimeter.
3. A regular octagon has 8 equal sides, each 6 cm long. What is its perimeter?
4. A trapezium has sides of 10 cm, 6 cm, 8 cm and 5 cm. Find its perimeter.
5. A parallelogram has sides of length 15 m and 9 m. What is its perimeter?
6. A circle has a diameter of 14 cm. Taking $\pi = \frac{22}{7}$, find its circumference.

Check your work

Attempt every question before you read this page.

Meet the Plane Shapes: Types and Properties

1. Trapezium (one pair of parallel sides) and parallelogram (both pairs of opposite sides parallel, no right angles).

   A trapezium is defined by having exactly one pair of parallel sides. A parallelogram is defined by having both pairs of opposite sides parallel; because its angles are not right angles, it is not a rectangle or square.
2. Similarity: both have four sides and four right angles. Difference: a square has all four sides equal, but a rectangle only has its opposite sides equal.

   Look at the shared properties first — both are quadrilaterals with four right angles. Then look for what differs: in a square every side is the same length, while a rectangle needs only opposite sides equal.
3. It is a circle, and the fixed distance is its radius.

   The only shape with no straight sides and a constant distance from a centre is a circle. The segment from the centre to the curve is the radius; a line through the centre across the whole circle is the diameter, which is twice the radius.
4. A hexagon has 6 sides and 6 angles, which is three more sides than a triangle's 3.

   A hexagon is named for its six sides, so it has 6 angles too. A triangle has 3 sides, so 6 − 3 = 3 more sides.

Getting to Know the Circle

1. a. 18 cm (d = 2 × 9).

b. 15 cm (r = 30 ÷ 2).
c. The circumference.
d. The diameter.
e. C = 2πr = 2 × 3.14 × 10 = 62.8 cm.
f. A sector.
g. Diameter = 2 × radius = 2 × 9 = 18 cm.
h. Radius = diameter ÷ 2 = 30 ÷ 2 = 15 cm.
i. The distance all the way around the edge is the circumference.
j. A chord through the centre is the diameter (the longest chord).
k. C = 2πr = 2 × 3.14 × 10 = 62.8 cm.
l. The region between two radii and the arc is the sector.

Finding the Perimeter of Plane Shapes

1. 36 m

   A square has four equal sides, so perimeter $= 4 \times 9 = 36$ m.
2. 38 cm

   Perimeter $= 2(\text{length} + \text{width}) = 2(12 + 7) = 2 \times 19 = 38$ cm.
3. 48 cm

   A regular octagon has 8 equal sides, so perimeter $= 8 \times 6 = 48$ cm.
4. 29 cm

   Add all four sides: $10 + 6 + 8 + 5 = 29$ cm.
5. 48 m

   Opposite sides are equal, so perimeter $= 2(15 + 9) = 2 \times 24 = 48$ m.
6. 44 cm

   Circumference $= \pi d = \frac{22}{7} \times 14 = 44$ cm.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-8fb095df	**Date**	
Subtopic	Plane shapes: types, properties and perimeter	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____